

4.3 Network Security

In this course, I will:

- Examine how to use network segmentation, protocols, services and how to harden a network environment and monitor its security
- Explore AWS network security tools used to manage, monitor, and protect AWS services, and take an in-depth look at using AWS CloudTrail, AWS CloudWatch, and AWS Elastic Load Balancing
- Practice creating an AWS CloudTrail, a network load balancer, and using AWS CloudWatch

Secure Cloud Networks Using Network Segmentation



Network Segmentation



Used to divide networks into multiple subnetworks



Enables better control of flow of traffic



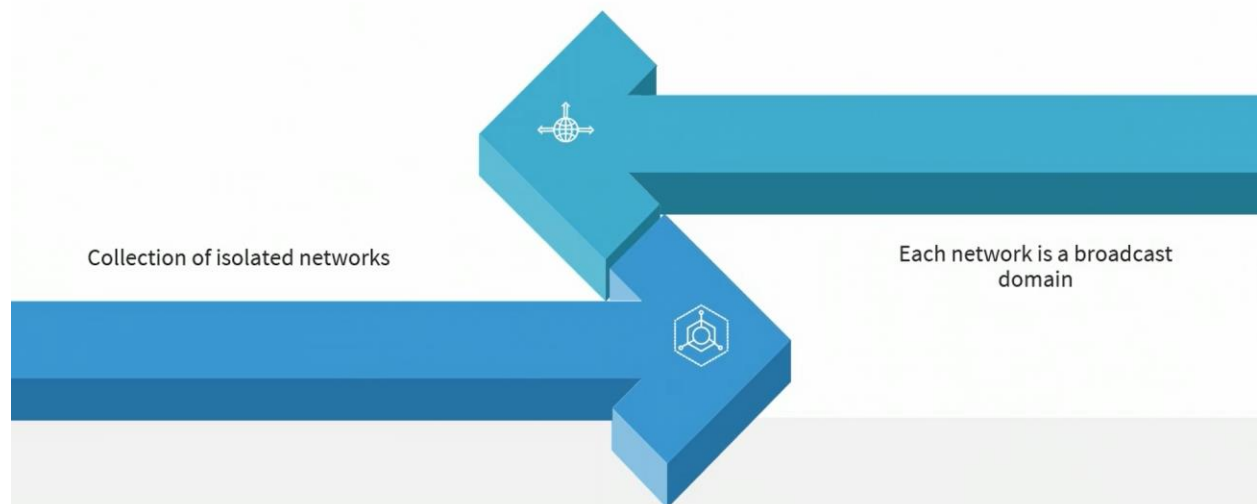
to control the flow of traffic,

Benefits of Network Segmentation



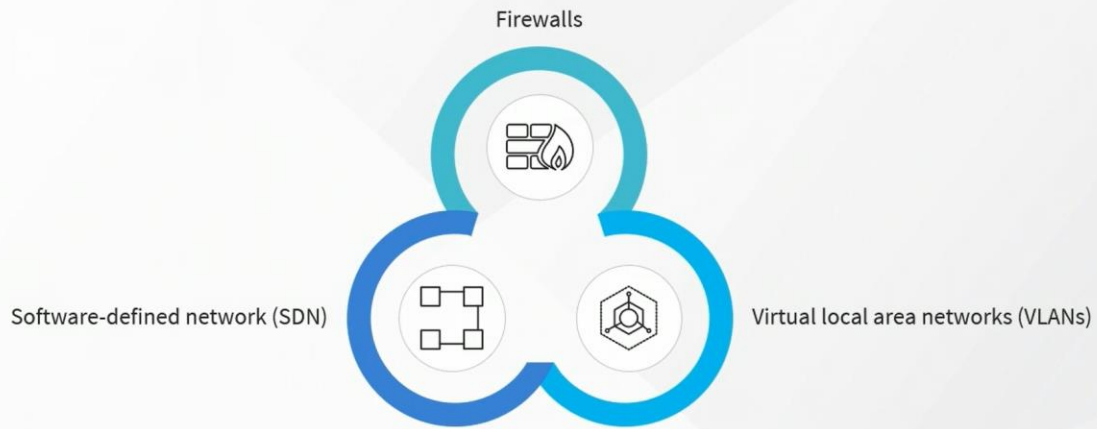
by having my services organized in different subnetworks.

Virtual Local Area Network (VLAN)



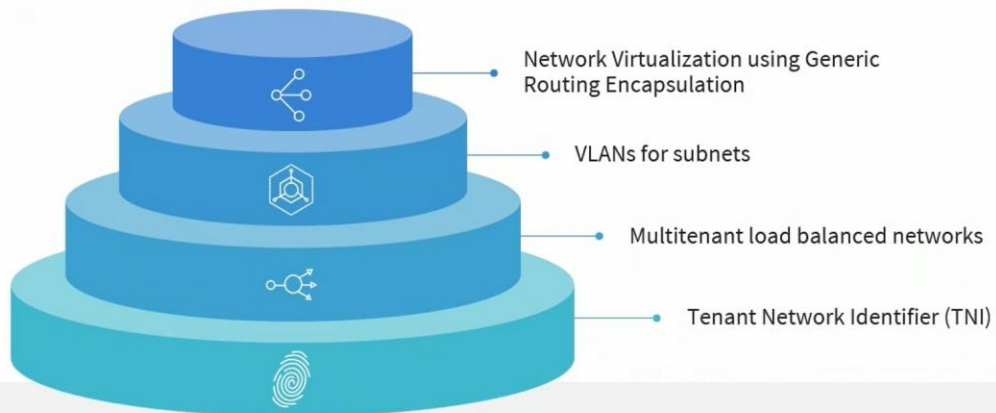
And what this means is that we can broadcast to a domain.

Segmenting Networks



but the idea is that I can define the network

NVGRE



And then finally, I can have something known as the



Defense-in-Depth



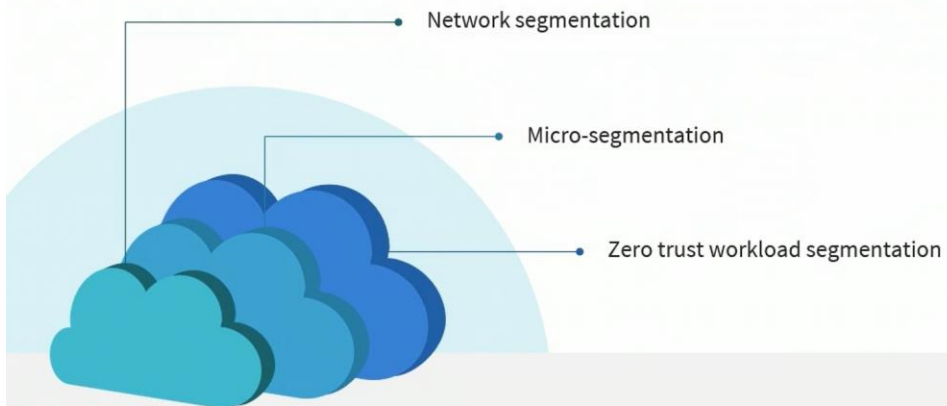
Prevent the spread of malware



Prevent unwanted network access

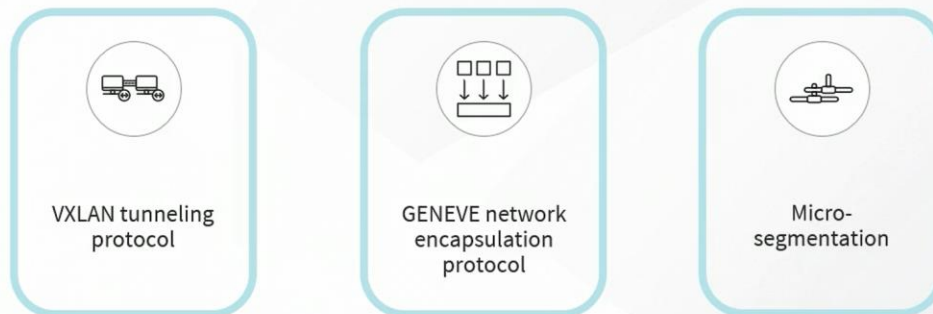
and it can also block hackers or unwanted network access,

Cloud Workload Segmentation



Finally, there's the zero trust cybersecurity model.

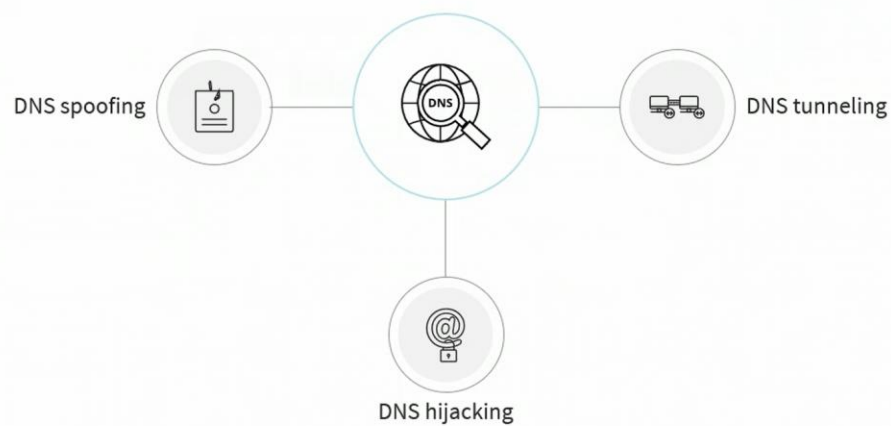
Other Segmentation Options



you'll be able to communicate with the outer layer.

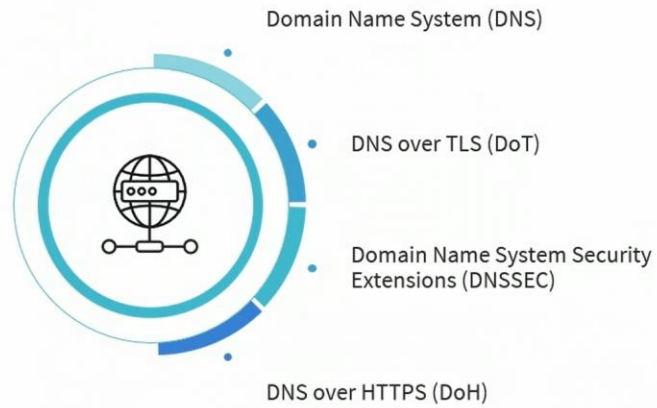
Secure Cloud Networks Using Network Protocols

DNS Security



to a completely different domain name server,

DNS Strategy



which is really similar to the DNS over TLS.

Time-based Protection



which is a beefier, a more newer way to handle things.

Tunneling



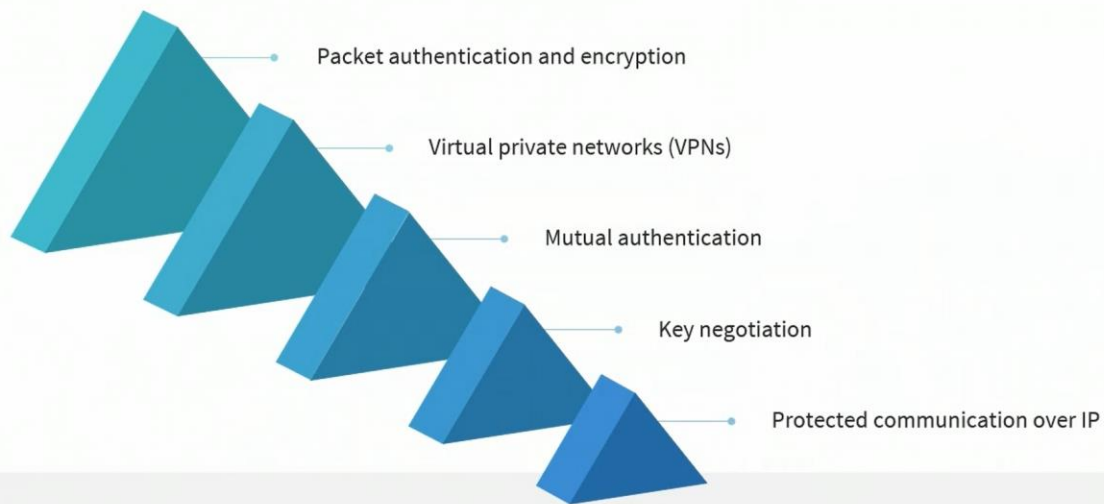
And this will do is it'll ensure secure communication.

Tunneling Protocols



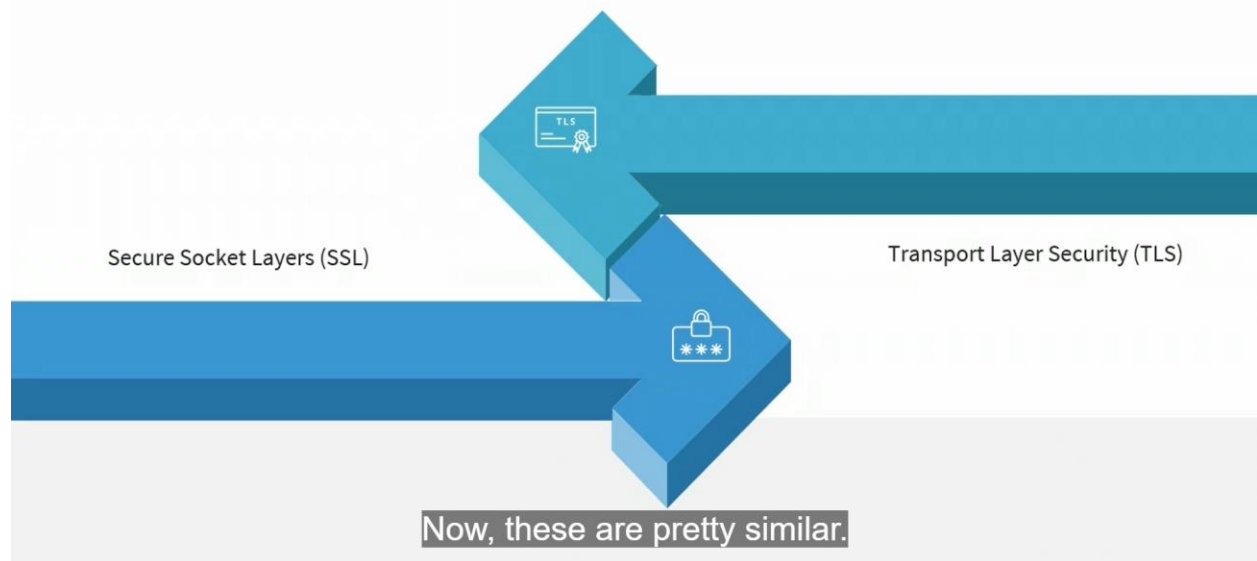
And then there's the Generic Routing Encapsulation or GRE.

Internet Protocol Security (IPsec)

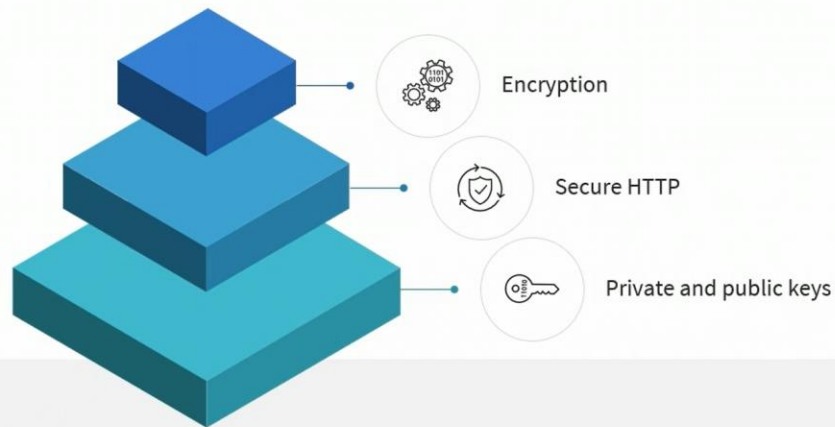


protected communications over the IP address by using the IPsec protocol

Secure Connections



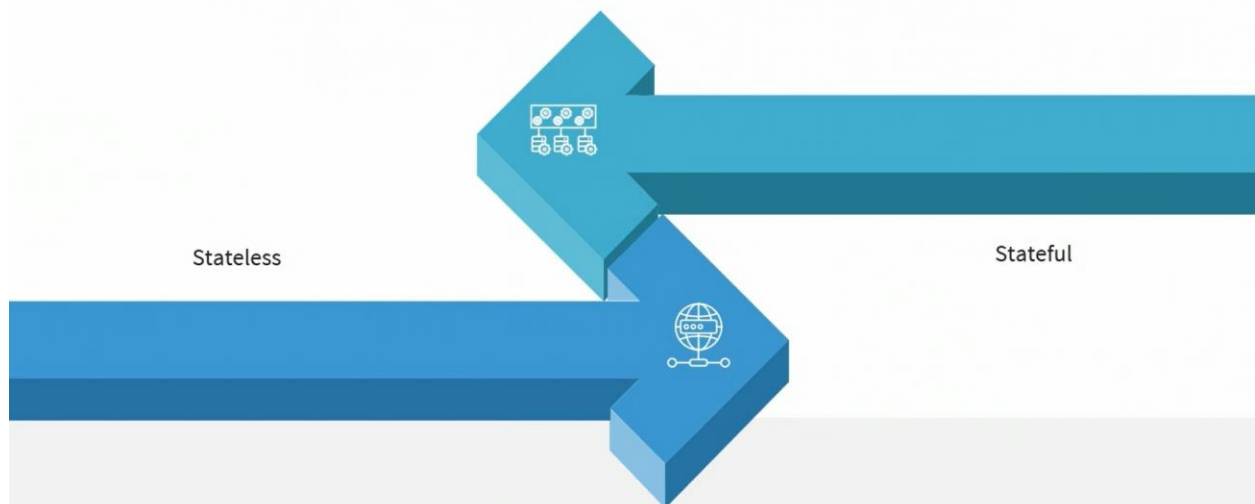
Hypertext Transfer Protocol Secure (HTTPS)



So, those are all the different types of network protocols

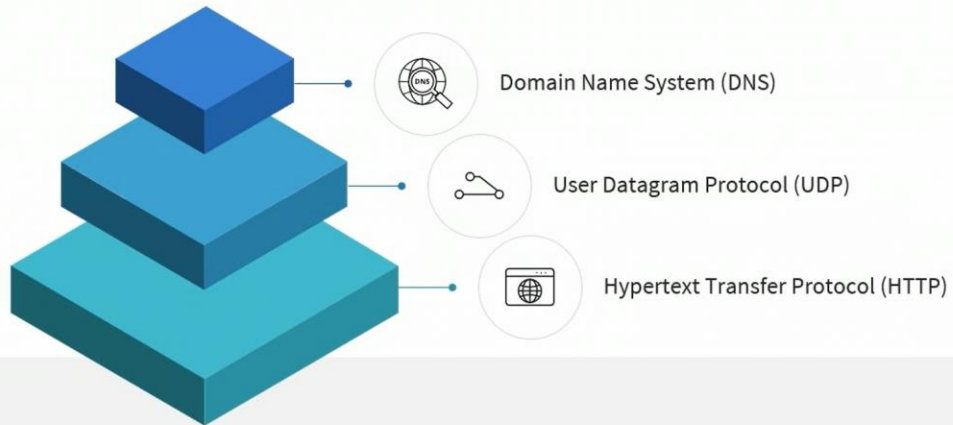
Secure Cloud Networks Using Network Services

Network Service Terminology



and the cookie is going to be sent to the stateful server,

Stateless Protocols



and they're used to simplify the overall design of a given server.



Stateful Protocols



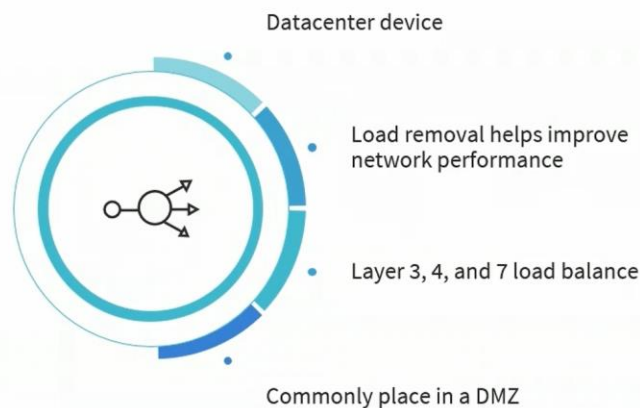
which is used for general networking management of a stateful application.

Web Application Firewall (WAF)



You're going to have to use things that will increase your cybersecurity.

Application Delivery Controller (ADC)



and it's commonly placed in between the web application firewall

Cloud DLP



and it's going to have to de-identify our data.

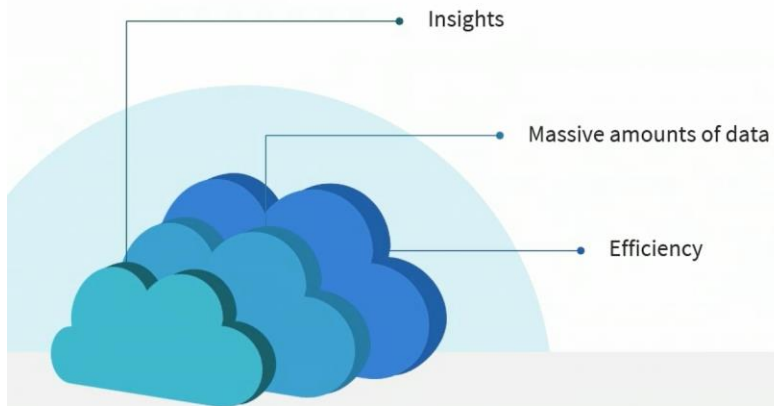
Network Packet Broker (NPB)



and networking principles in stateful and stateless applications.

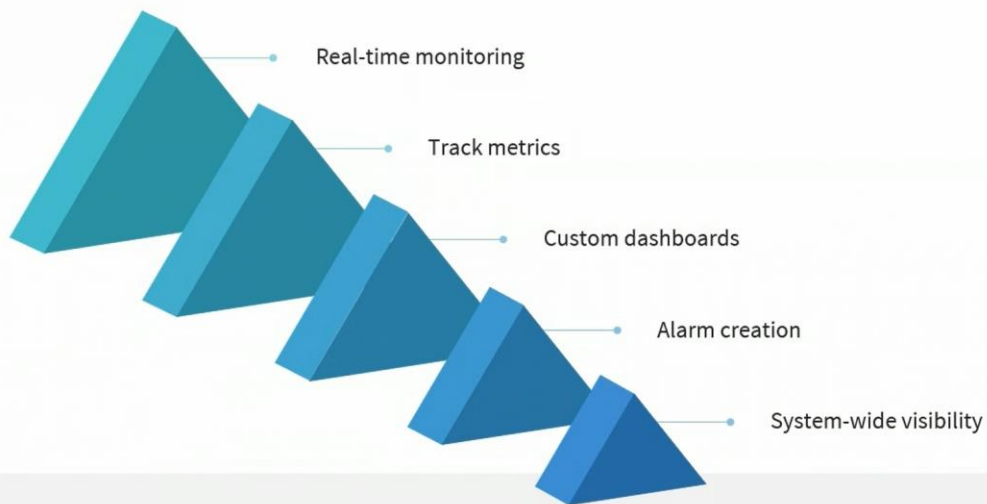
Network Flow Analysis and Event Monitoring

Cloud Logging



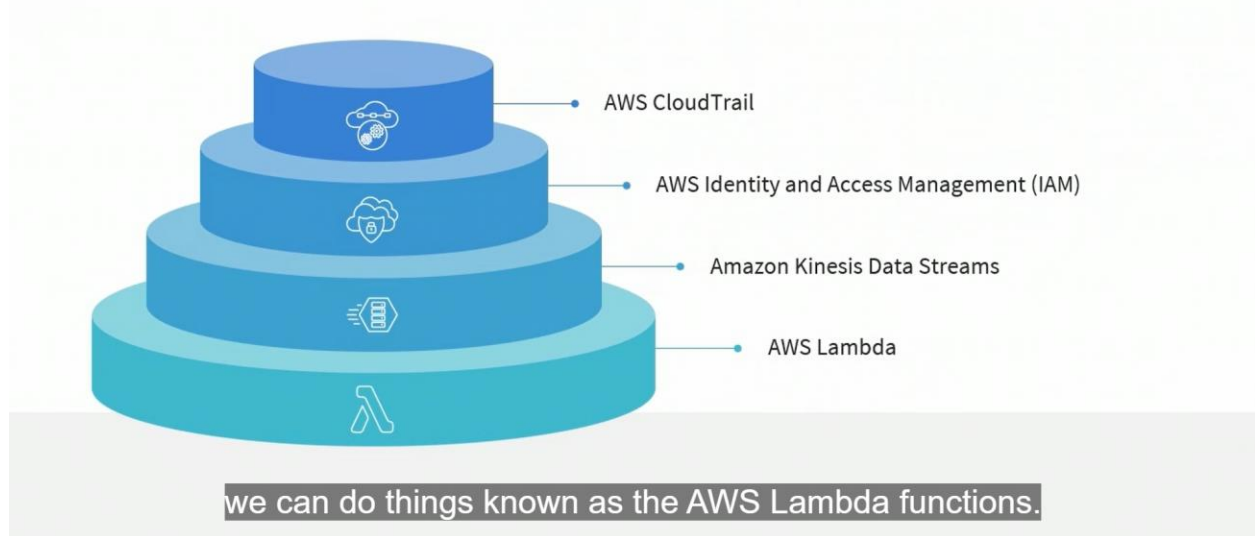
by using a cloud provider to have some type of logging, scanning

Amazon CloudWatch

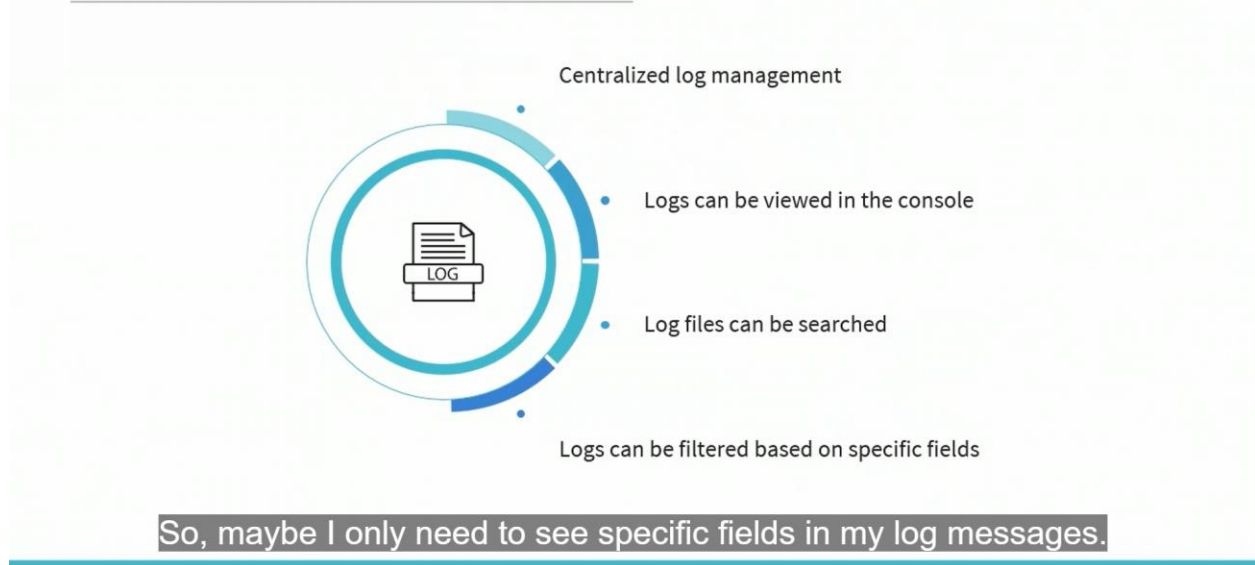


This provides system-wide visibility across all of our regions

Amazon CloudWatch Logs and AWS Services



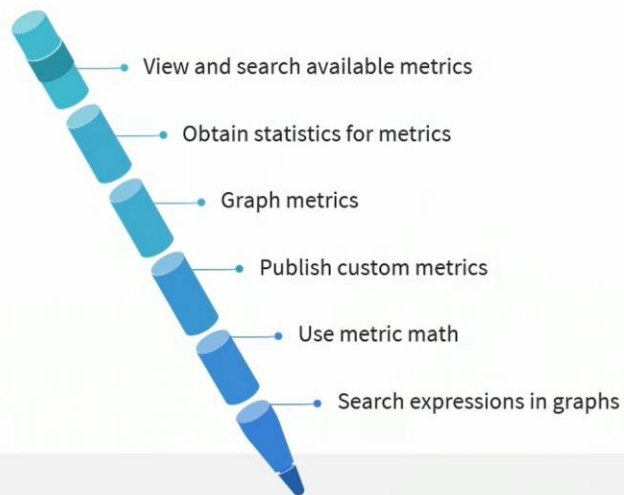
Amazon CloudWatch Logs



Amazon CloudWatch Logs



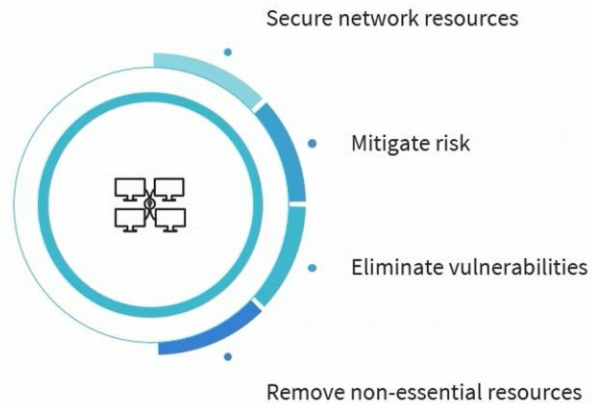
Amazon CloudWatch Metrics



look at certain pieces of logs that have been generated by one of my services.

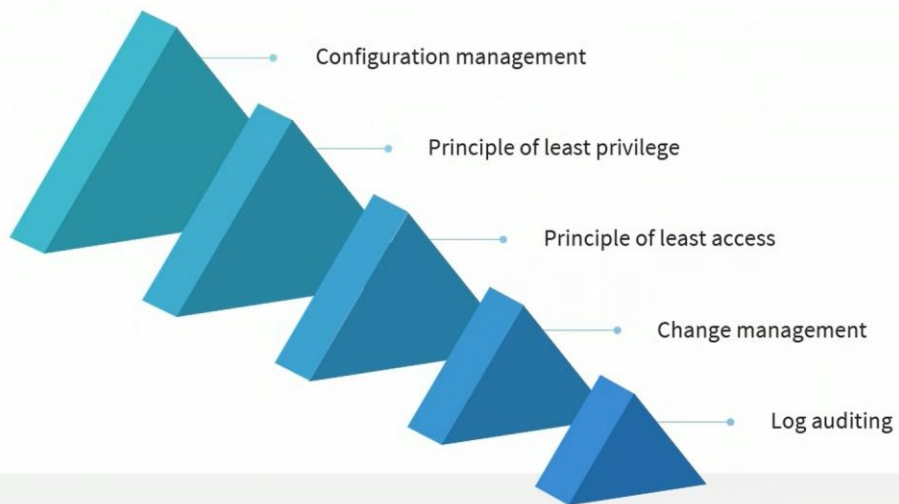
Network Hardening

Network Hardening



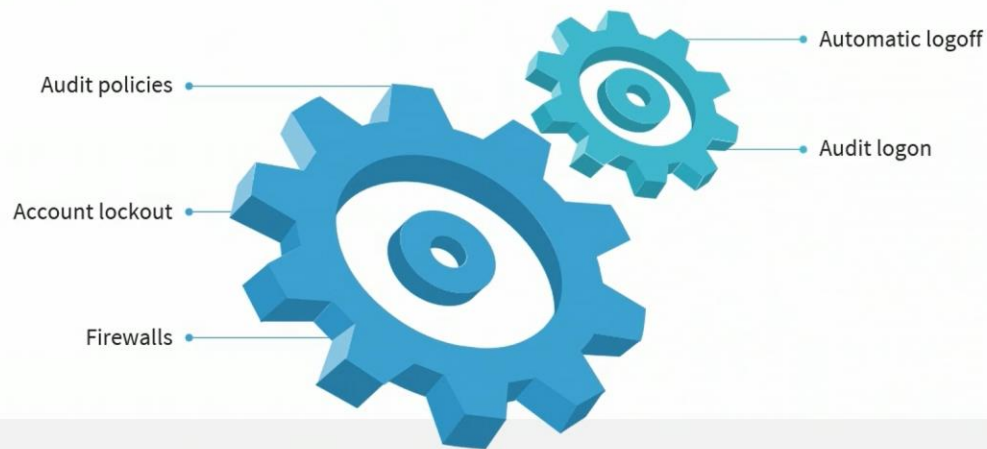
We can also remove any non-essential resources,

The Cornerstone of Network Hardening



or that's attempting to have access to our network will allow us

Best Practices for Hardening



And then there's the audit logon and the audit logons is allowing you to

Best Practices for Hardening



Close open ports



Synchronize with NTP



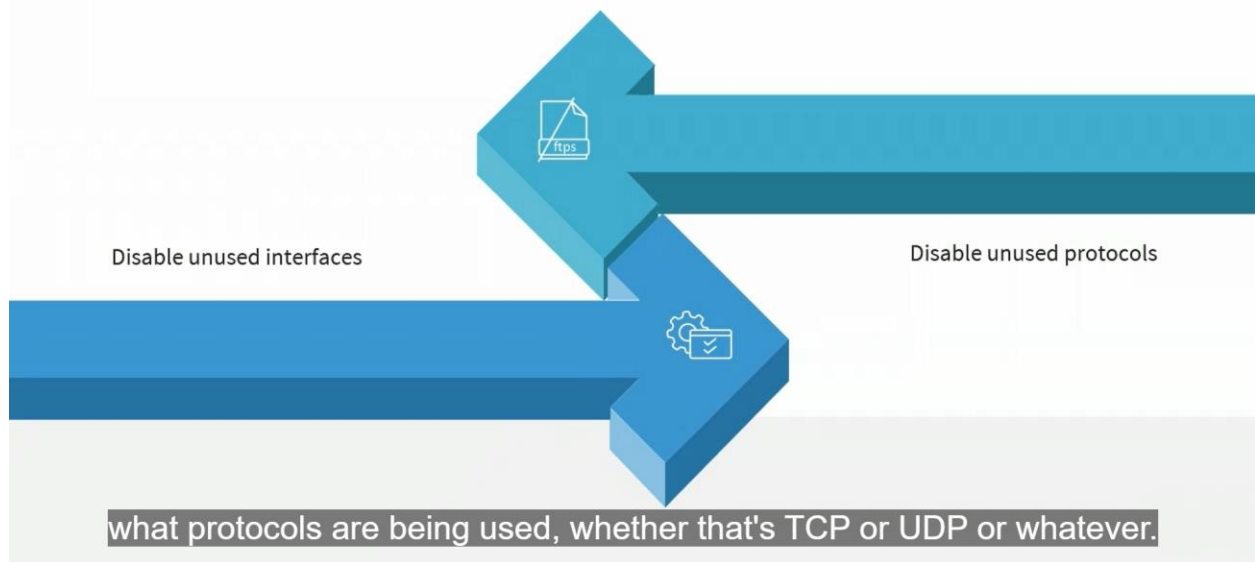
Password policies



Auditing and logging

and then I'll have any auditing and logging systems in place.

Non-essential Resources



Cloud Hardening



And finally, you want to utilize private subnets,

Cloud Hardening Solutions



Now, what I mean by layered is that certain things

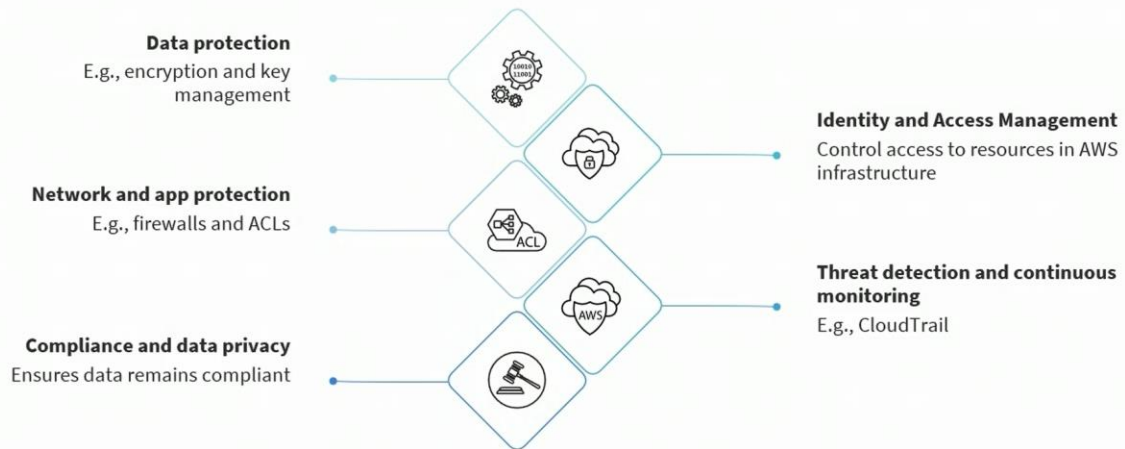
Managed Detection and Response (MDR)



identify what threat is going to be attacking my VPC

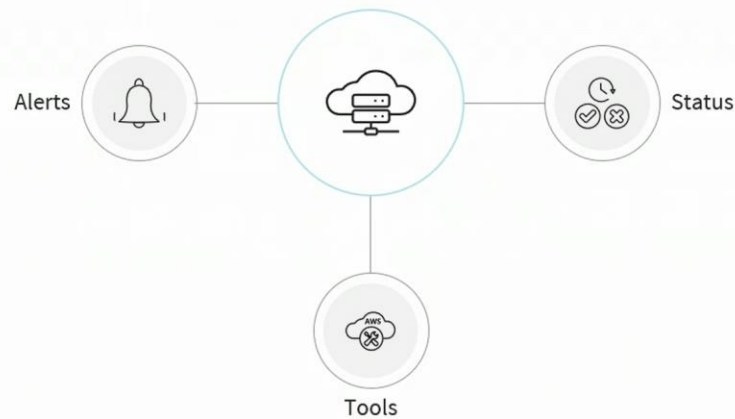
AWS Network Security Tools

AWS Security Tools



there's things that it will ensure our data will remain compliant.

AWS Security Hub

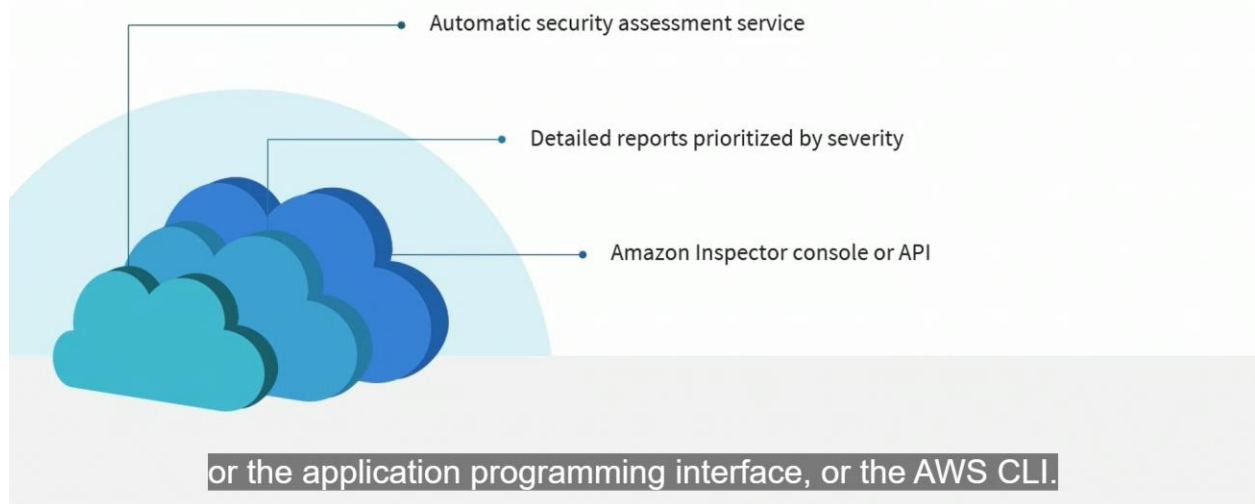


to manage those occurrences.

Amazon GuardDuty



Amazon Inspector

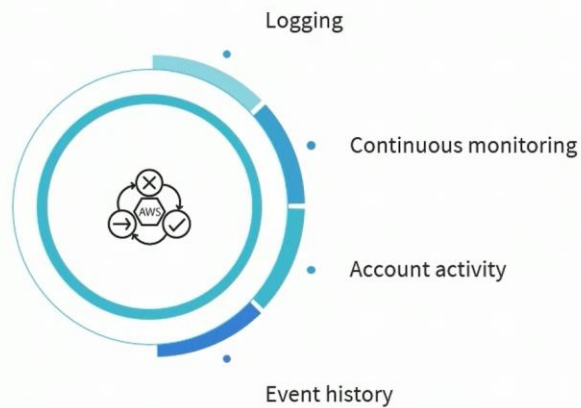


AWS Config



over what these resource configurations are doing.

AWS CloudTrail



It will also provide event history so you can go back in time

AWS Security Services



And then there's the AWS IoT Device Defender that will protect

AWS Security Management Tools



we can analyze all of those logs in one's

AWS CloudTrail

AWS CloudTrail

Risk auditing



Governance



Compliance



and compliance of our AWS account.

CloudTrail Trails



Archive



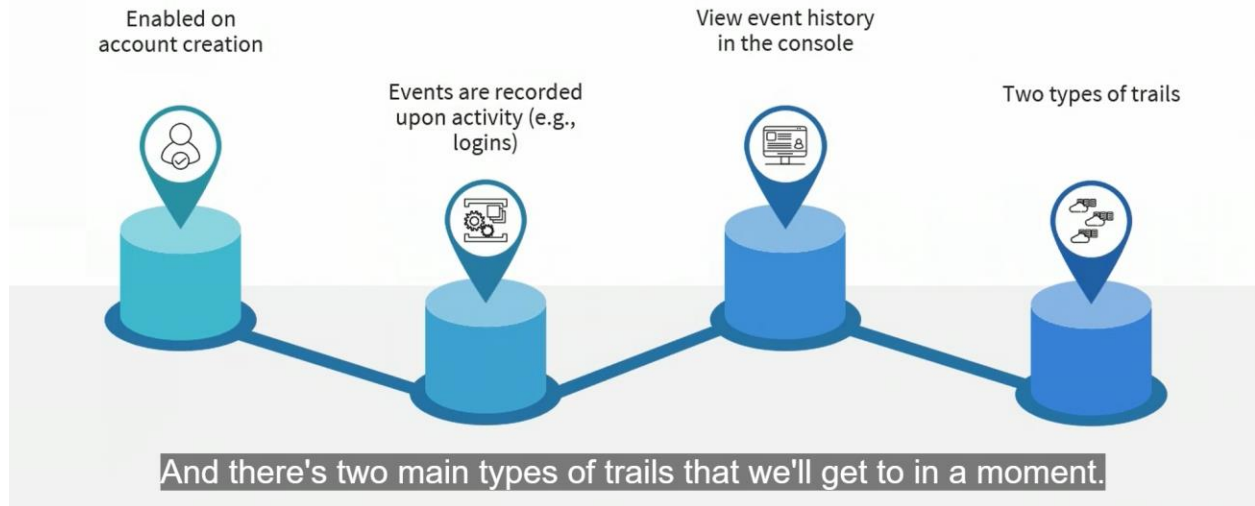
Analyze



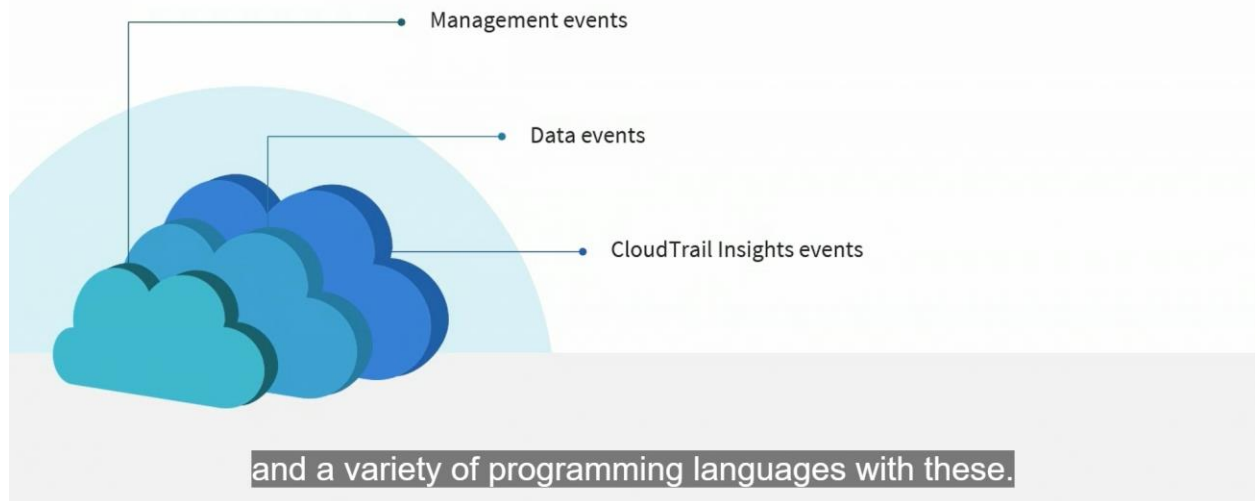
Respond

such as a malicious actor trying to access one of our AWS services.

How AWS CloudTrail Works



CloudTrail Events



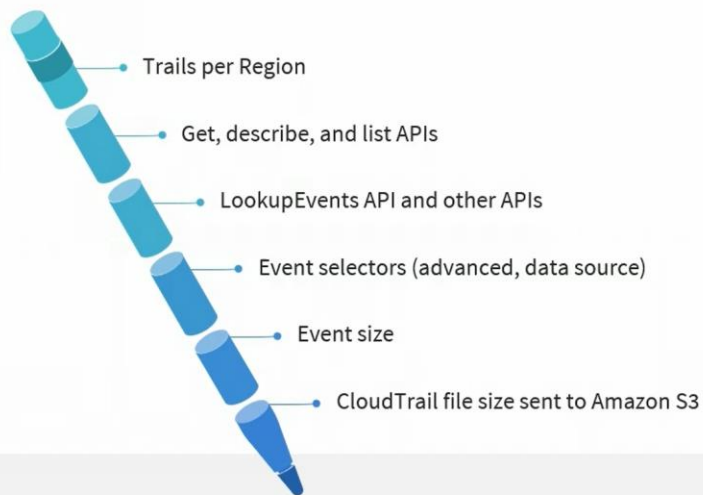
CloudTrail Log File



- A log file contains one or more records
- CloudTrail uses a specific log file name format
- AccountID_CloudTrail_RegionName_YYYYMMDDTHHmmZ_UniqueString.FileNameFormat

you can download and analyze yourself on your local machine if you'd like.

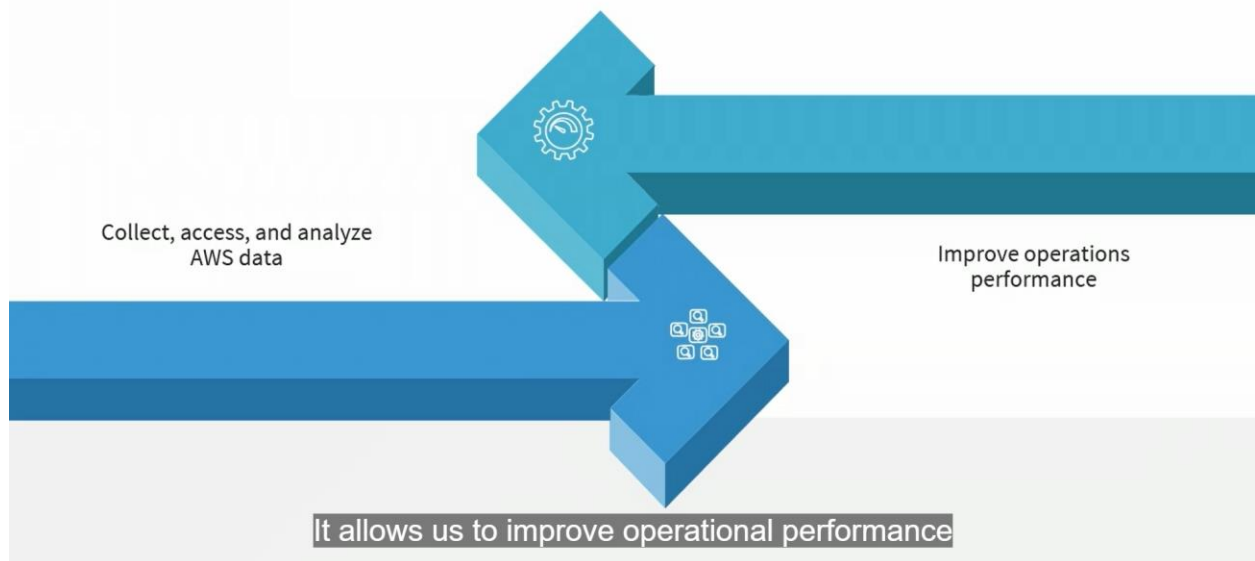
AWS CloudTrail Limitations



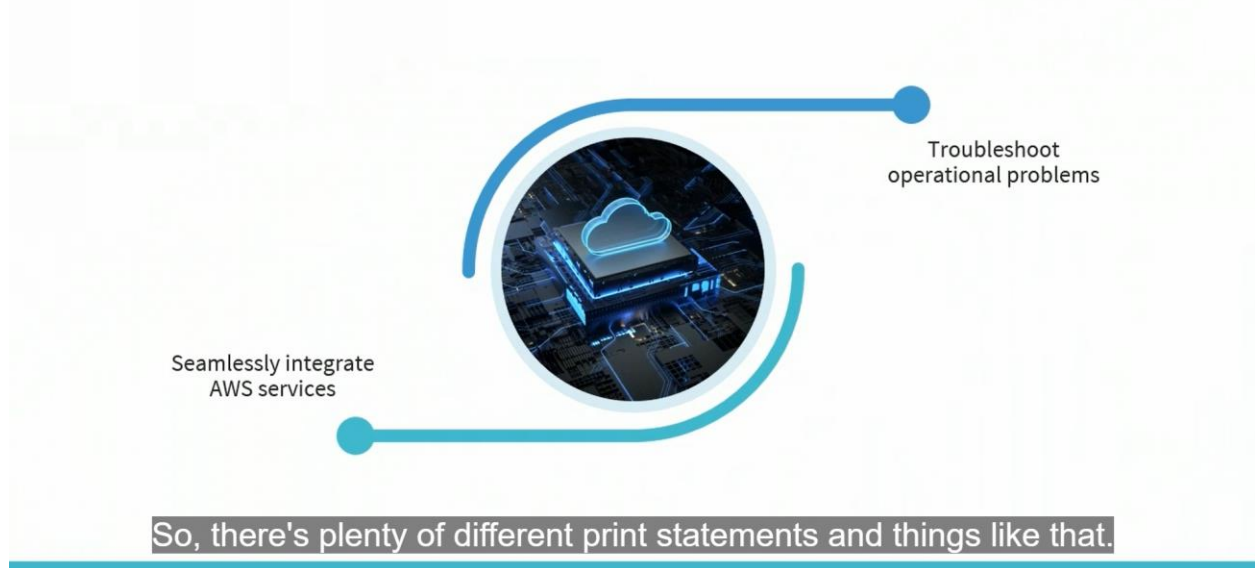
And you can have CloudTrail file size that's sent to the Amazon S3.

AWS CloudWatch

Amazon CloudWatch



Amazon CloudWatch



Integrating Services with CloudWatch



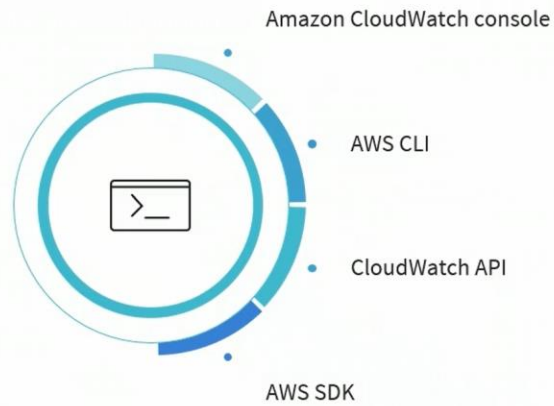
There's things like AWS CloudTrail and AWS Identity and Access Management

Amazon CloudWatch Process



Finally, we get to analyze all of this information

Amazon CloudWatch Access



library for a particular programming language that you'd like to use.

Use Cases



an analysis to see how your cost is either increasing or decreasing

CloudWatch Alarms



Metric alarms



Composite alarms



it's three or four things happening in quick succession across

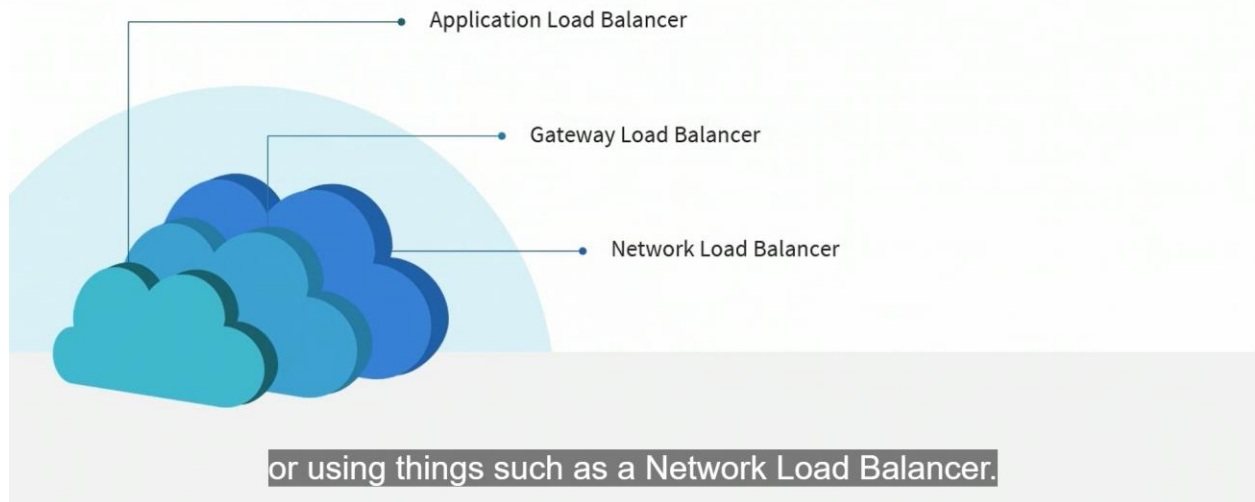
AWS Elastic Load Balancer (ELB)

Elastic Load Balancing (ELB)

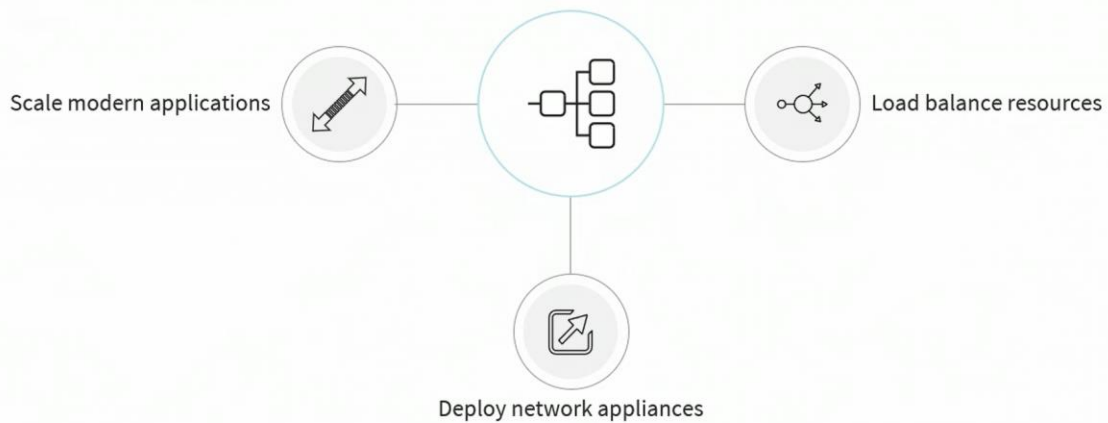


is you can monitor the health and performance

Elastic Load Balancing (ELB)

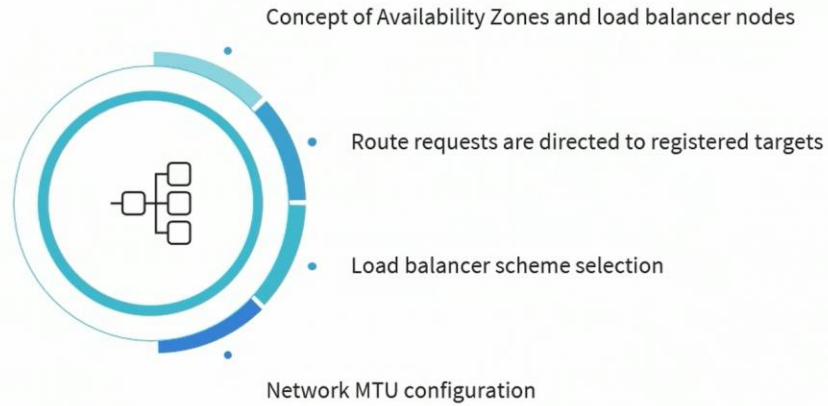


Common Use Cases



I can deploy my network appliances from vendor

How Elastic Load Balancing Works



such as what is going to be accepted,

Working with Elastic Load Balancing



and we get to incorporate Identity and Access Management

Working with Elastic Load Balancing



And finally, I can have interface our VPC endpoints. So, that

Creating an AWS CloudTrail

Configuring AWS CloudWatch for Real-time Monitoring

Creating a Network Load Balancer in AWS

1. Which AWS service is used for real-time monitoring?
 - AWS CloudWatch
2. What two connection types are associated with secure networks?
 - SSL
 - TLS
3. Which definition correctly describes the principle of least privilege?
 - Control of roles/permissions of users/devices so they only have enough permissions to do their tasks
4. Which feature represents the group of machines that a load balancer will forward requests to?
 - Target group

5. Which AWS service is used for risk auditing in the cloud?
 - AWS CloudTrail
6. Which AWS service is used to manage and monitor configurations for AWS services?
 - AWS Config
7. Which system or service routes a domain name to an IP address?
 - DNS
8. What is network segmentation used for?
 - To divide networks into multiple smaller networks
9. Which AWS CloudWatch feature sends an alert when a single metric reaches a threshold?
 - Metric alarms
10. Which dashboard type is always created in AWS CloudWatch whenever a new service is used?
 - Automatic dashboard
11. Which AWS service manages encryption keys when encrypting generated data from AWS CloudTrail?
 - AWS KMS
12. Which AWS offering functions as a middleman to forward requests to target servers?
 - Elastic Load Balancer